AI-POWERED SUPPLY CHAIN DISRUPTION ASSISTANT

Evaluation Report

ML Model Performance, Retrieval Quality, and RAG Assessment

Dataset: 100,000 records | Test split: 15,000 records

RAGAS benchmark: 10 test cases | 9/10 passed (90.0%)

Offline retrieval benchmark: 5 cases | All PASS

1. Evaluation Objectives

This report presents all measured evaluation results for the AI-Powered Supply Chain Disruption Assistant across three independent dimensions:

- Offline ML model performance — held-out test metrics for four trained Random Forest models (evaluation/model_metrics.json).
- Retrieval quality — offline benchmark of the hybrid Chroma + BM25 retriever (evaluation/reports/latest_offline_evaluation.json).
- RAG output quality — RAGAS faithfulness, answer relevancy, context precision, context recall (evaluation/reports/latest_evaluation.json).

All metrics are sourced directly from recorded artifacts. Where an evaluation was not performed or results were unavailable, this is stated explicitly. No metrics have been fabricated or extrapolated.

2. Evaluation Methodology

Offline ML Evaluation

scripts/train_models.py splits 100,000 records 70/15/15. Validation metrics guide model selection (Random Forest vs Extra Trees). Test metrics are measured on the 15,000 held-out rows. Results are saved to evaluation/model_metrics.json with a training timestamp (2026-10-05T20:28:32Z).

Offline Retrieval Benchmark

evaluation/evaluate.py runs 5 labeled test cases from evaluation/cases.json against the application's Chroma and BM25 indexes (top-K = 10). Metrics: context precision@10, labeled recall@10, mean reciprocal rank (MRR), recommendation support rate, evidence consistency rate, and empty-result handling. Results are written to evaluation/reports/latest_offline_evaluation.json. No MySQL or LLM API key required.

RAGAS Online Evaluation

POST /api/disruptions/analyze automatically triggers RAGAS evaluation after every analysis. A 10-test benchmark was executed separately (timestamp 2026-10-08T17:26:34Z, results in evaluation/reports/latest_evaluation.json) using RAGAS (faithfulness, answer relevancy, context precision, context recall). Threshold for passing each test: 0.75 (75%). Overall score = mean of four RAGAS metrics per test case.

3. Dataset Split

Split	Rows	Percentage	Purpose
Training	70,000	70%	Model fitting.
Validation	15,000	15%	Model selection by MAE (regression) or F1 (classification).
Test (held-out)	15,000	15%	Final metric measurement; unseen during training and validation.

Features: disruption_type, industry, supplier_tier, supplier_region, supplier_size, has_backup_supplier, disruption_severity (7 features). Post-event and outcome columns excluded to prevent label leakage.

4. ML Regression Results

Task 1 — Production Impact (production_impact_pct)

Metric	Validation	Test (Held-Out)
Algorithm	Random Forest Regressor	Random Forest Regressor
MAE	8.05 percentage points	7.99 percentage points
RMSE	—	9.92 percentage points
R squared	—	0.684

Interpretation: Average model estimate differs by ~8 pp from actual. R2 of 0.684 explains ~68% of production impact variance.

Task 2 — Revenue Loss (revenue_loss_usd)

Metric	Validation	Test (Held-Out)
Algorithm	Random Forest Regressor	Random Forest Regressor
MAE	$1,740,869	$1,774,595
RMSE	—	$4,419,908
R squared	—	0.374

Interpretation: Revenue loss is highly right-skewed ($10,000–$373 million; median $875,466). R2 = 0.374 indicates moderate accuracy. Predictions are directional estimates requiring human validation for high-stakes financial decisions.

Task 3 — Full Recovery Time (full_recovery_days)

Metric	Validation	Test (Held-Out)
Algorithm	Random Forest Regressor	Random Forest Regressor
MAE	32.25 days	31.74 days
RMSE	—	45.21 days
R squared	—	0.369

Interpretation: Recovery time ranges 8–798 days (median 61 days). MAE of 31.74 days and R2 0.369 indicate moderate accuracy. Wide range in training data contributes to variance.

5. ML Classification Results

Task 4 — Permanent Supplier Change (permanent_supplier_change)

Metric	Validation	Test (Held-Out)
Algorithm	Random Forest Classifier	Random Forest Classifier
F1 Score	0.418	0.401
Accuracy	—	0.640 (64.0%)
Precision	—	0.332
Recall	—	0.507
ROC-AUC	—	0.635

Class distribution: 23,920 True (24%) vs 76,080 False (76%). F1 = 0.401 on minority class. Precision 0.332 means ~33% of predicted positives are correct. Recall 0.507 catches ~51% of actual supplier changes. ROC-AUC 0.635 shows moderate discriminative ability above chance (0.5).

6. Retrieval Evaluation (Offline Benchmark)

Source: evaluation/reports/latest_offline_evaluation.json. Evaluation mode: application hybrid retriever (Chroma + BM25). Top-K: 10. Test cases: 5.

Metric	Score	Threshold	Status	Description
Context Precision@10	1.00	0.95	PASS	Share of retrieved cases satisfying test-case labeled business filters.
Labeled Recall@10	1.00	0.75	PASS	Share of hand-labeled reference cases retrieved in top-10.
Mean Reciprocal Rank	1.00	0.75	PASS	Rank of the first hand-labeled reference case.
Recommendation Support Rate	1.00	1.00	PASS	Recommended strategy belongs to retrieved historical response groups.
Evidence Consistency Rate	1.00	1.00	PASS	Rationale values and cited case IDs match retrieved evidence.
Empty-Result Handling	1.00	1.00	PASS	No-match query yields no cases and an insufficient-evidence recommendation.
DeepEval Faithfulness	N/A	N/A	NOT EVALUATED	LLM judge not run in this offline pass.
DeepEval Answer Relevancy	N/A	N/A	NOT EVALUATED	LLM judge not run in this offline pass.
DeepEval Contextual Relevancy	N/A	N/A	NOT EVALUATED	LLM judge not run in this offline pass.

Overall status: PASS. DeepEval metrics were not evaluated and are reported as Not Evaluated (not zero).

7. RAGAS Benchmark (10 Test Cases)

Source: evaluation/reports/latest_evaluation.json. Timestamp: 2026-10-08T17:26:34Z.

Summary

Property	Value
Total test cases	10
Passed (score >= 75%)	9
Failed (score < 75%)	1 (TC-04)
Pass rate	90.0%
Average score (all 10 cases)	84.6%
Overall benchmark status	PASSED

Per-Test Results

ID	Disruption Type	Sev	Score	Status	Key Finding
TC-01	Port Congestion	4	97.5%	EXCELLENT	All 4 RAGAS metrics passed including faithfulness 100%.
TC-02	Factory Incident	5	82.5%	GOOD	Faithfulness 50% (ungrounded action specifics); other metrics passed.
TC-03	Cyber Attack	5	82.5%	GOOD	Faithfulness 50% (ungrounded mitigation specifics); other metrics passed.
TC-04	Logistics Delay	1	67.5%	FAILED	Faithfulness 50%, Answer Relevancy 20% (disproportionate response for severity 1).
TC-05	Factory Incident	4	82.5%	GOOD	Faithfulness 50% (mixed-context grounding); other metrics passed.
TC-06	Factory Incident	5	83.8%	GOOD	Faithfulness 50% (mixed-context actions); other metrics passed.
TC-07	Natural Disaster	4	82.5%	GOOD	Faithfulness 50% (ungrounded specifics); other metrics passed.
TC-08	Labor Strike	3	85.0%	EXCELLENT	Faithfulness 50%; Answer Relevancy 90%; Context metrics 100%.
TC-09	Supplier Bankruptcy	4	97.5%	EXCELLENT	All metrics passed; Faithfulness 100%, Answer Relevancy 90%.
TC-10	Regulatory Block	3	85.0%	EXCELLENT	Faithfulness 50% (mixed context); other metrics passed.

RAGAS Metric Aggregates

Metric	Score Range	Notes
Faithfulness	50%–100%	TC-01 and TC-09: 100%. 8/10 cases: 50% due to LLM generalizing beyond a single retrieved context document.
Answer Relevancy	20%–90%	TC-04: 20% (disproportionate response for severity 1 event). All other cases: 80%–90%.
Context Precision	100% all 10 cases	Retriever consistently ranked relevant cases at the top position.
Context Recall	100% all 10 cases	All 5 query facets (type, industry, region, tier, severity) covered in retrieved context.

8. Limitations

- Faithfulness scores: 8/10 cases scored 50% because the LLM recommendation includes action specifics (e.g., 'reroute freight') not directly cited in a single retrieved context. This reflects LLM generalization beyond retrieved evidence, not retrieval failure.
- TC-04 failure: answer relevancy 20% indicates the system may over-respond to low-severity events. This is a documented limitation for improvement.
- Revenue loss regressor (R2 0.374) and recovery time regressor (R2 0.369) have moderate accuracy; unsuitable for exact financial planning.
- Permanent supplier change classifier (precision 0.332) produces a high false-positive rate.
- DeepEval judge metrics from the offline benchmark were not run and must be reported as Not Evaluated, not zero.
- All metrics are measured on locally available dataset and retrieval artifacts; results may differ if rebuilt.

9. Conclusion

The system achieves strong retrieval performance (context precision and recall = 1.00, MRR = 1.00 across 5 offline benchmark cases). The RAGAS benchmark demonstrates 90% pass rate with average score 84.6%, indicating generally good RAG quality. Faithfulness is the primary improvement area — the LLM recommendation engine sometimes introduces action specifics not directly traceable to a single retrieved context document. Production impact prediction (R2 0.684) is the strongest ML model; revenue loss and recovery time predictions have moderate accuracy suitable for directional guidance but not exact financial planning.